Ku-Yu Fan

Academic Curriculum Vitae

Ph.D. Candidate in Mathematics | Nagoya University

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27 OCT 2026

Research Interests

Multiple zeta values; motivic periods and coactions; Schur multiple zeta values; arborified multiple zeta values; p-adic multiple zeta values; multiple polylogarithms; p-adic multiple L-functions.

Education

2024–Present
Ph.D. in Mathematics | Nagoya University
Graduate School of Mathematics. Expected completion: September 2027.
Advisor: Hidekazu Furusho

2021–2024
M.Sc. in Mathematics | National Taiwan University
Institute of Mathematics.
Advisor: Nobuo Sato

2017–2021
B.Sc. in Mathematics | National Tsing Hua University
Department of Mathematics.

Refereed Research Articles

2026
Coaction formula for the motivic version of Yamamoto's integral
Mathematische Zeitschrift ***314***, *Article 35 (2026).*
DOI: 10.1007/s00209-026-04138-w

2026
A map between arborifications of multiple zeta values
Journal of Algebra (2026). DOI: 10.1016/j.jalgebra.2026.06.006

Preprints

2026
Regularised double shuffle relations for planar Arborified Zeta Values
Pierre Catoire, Pierre J. Clavier and Ku-Yu Fan. arXiv:2609.02346.

2026
p-adic multiple L-functions and twisted multiple Bernoulli numbers
Ku-Yu Fan. arXiv:2607.04086.

2026
p-adic multiple zeta values of integer indices
Ku-Yu Fan. arXiv:2603.22923.

Expository Publication

2021
The Number of Lattice Lines in Lattice Figures [in Chinese]
Ku-Yu Fan and Feng-Mei Lin. Mathmedia (Shuxue Chuanbo) ***45****(2), 44–53 (2021).*
Article webpage

Selected Research Talks

Sep 2026 — **Explicit Formulas for Cyclotomic Multiple Bernoulli Numbers**
NCTS Number Theory Seminar, National Tsing Hua University.

Sep 2026 — **An explicit formula for the special values of the p-adic multiple L-functions**
Mathematical Society of Japan, Autumn Meeting, Kobe University.

Jul 2026 — **Double shuffle relations for p-adic multiple zeta values of integer indices**
5th International Workshop on Multiple Zeta Values and Related Topics, Central South University.

Jul 2026 — **Double shuffle relations for integer indices and their application to p-adic multiple zeta values**
25th Sendai–Hiroshima Number Theory Conference, Tohoku University.

Jun 2026 — **p-adic multiple zeta values and double shuffle relations for integer indices**
Algebraic and Combinatorial Perspectives in the Mathematical Sciences, online.

Mar 2026 — **p-adic multiple zeta values of integer indices**
Meeting in the Middle (Characteristic 0 and p Multiple Zeta Values), University of Hawaii at Manoa.

Feb 2026 — **p-adic multiple L-function and twisted multiple Bernoulli numbers**
MATHSCI Freshman Seminar, Nagoya University.

Sep 2025 — **A Map Between Arborifications of Multiple Zeta Values**
RIMS Symposium on Multiple Zeta Values, Kyoto University.

Sep 2025 — **A Map Between Arborifications of Multiple Zeta Values**
Mathematical Society of Japan, Autumn Meeting, Nagoya University.

Jul 2025 — **A Map Between Arborifications of Multiple Zeta Values**
4th International Workshop on Multiple Zeta Values, Tongji University.

Mar 2025 — **A map between arborifications of multiple zeta values**
Monthly Meeting, Nagoya University (blackboard talk).

Nov 2024 — **Coaction Formula for the Motivic Version of Yamamoto’s Integral**
2nd Kindai Workshop: Multiple Zeta Values and Modular Forms, Kindai University.

May 2024 — **Coaction formula for the motivic version of Yamamoto’s integral**
Monthly Meeting, online.

Teaching Experience

2025–2026 **Nagoya University | Teaching Assistant**
Special Topics in Foundations (Analysis) — Spring 2026.
Calculus — Spring and Fall 2025.

2021–2024 **National Taiwan University | Teaching Assistant**
Calculus — Fall 2022, Spring and Fall 2023, Spring 2024.
Modern Algebra — Fall 2022, Spring 2023.
Linear Algebra — Fall 2021, Spring 2022.

2020–2021 **National Tsing Hua University | Teaching Assistant**
Elementary Number Theory — Spring 2021.
Calculus — Fall 2020, Spring 2021.
Advanced Calculus — Spring 2020.

Fellowships and Awards

2024–2027 **Japan–Taiwan Exchange Association Scholarship**
Graduate Research Students Fellowship (October 2024–September 2027).
Programme information

2019 **Professor Han-Ching Lai Memorial Scholarship**
Department of Mathematics, National Tsing Hua University. Details.

Academic Service and Organization

2026 **Organizing Committee — Nagoya–Kansai Young Researchers' Zeta Meeting**
Joint meeting of the 19th Young Mathematicians Conference on Zeta Functions and the 70th Kansai Multiple Zeta Seminar; supported by a selected Nagoya University student project (2025).
Meeting webpage

Sep 2024– **Organizer and Representative — Monthly Meeting**
Monthly research meeting on multiple zeta values. Meeting webpage.

Selected Patents

2021 **Single stacking element and component structures for multi-style combinations**
Taiwan utility model M612210.

2018 **Rhombic dodecahedron puzzle and multiple rhombic dodecahedron puzzle**
Taiwan utility model M566100; US Patent 10,722,782.

2017 **Hexagonal prismatic packing puzzle**
Taiwan utility model M548578; US Patent 10,322,337.

All listed patents co-invented with Wen-Liang Hung.